
Bài 7 – Phân tích và Quản lý đầu tư nâng cao

Thực hành Phân tích hồi qui

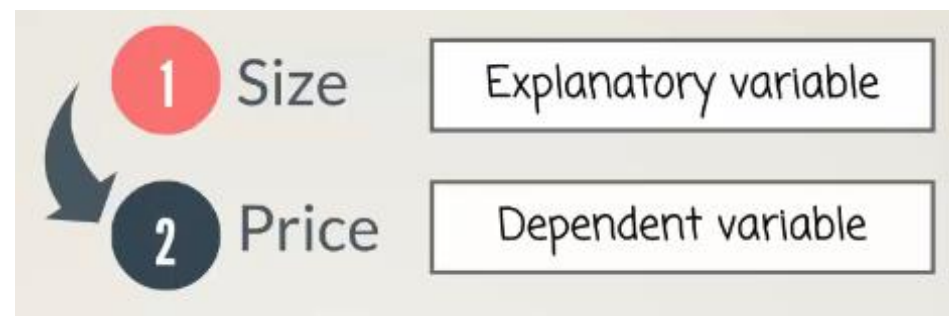
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Regression Analysis

Regression Analysis

- **Regression Analysis:** Quantifying the relationship between two variables
- **Simple Regression:** using only one variable

The larger house, the higher its price....



- **Multivariate regression:** using more than one variable

Simple Regression

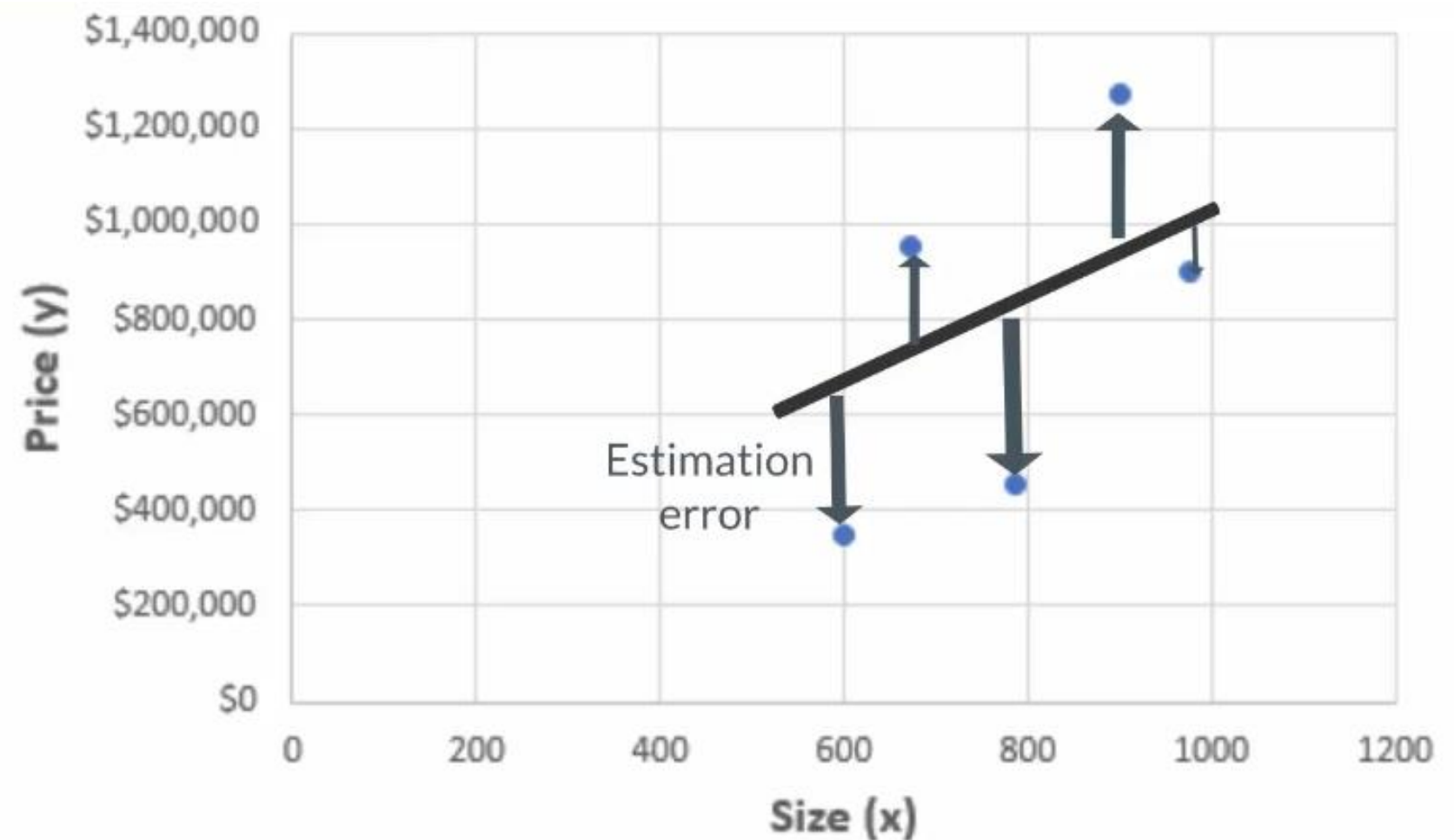
y = house price
x = house size

(1) How do we draw the best-fitting line?

(2) The best fitting line contains the least amount of estimation error.

$$y = \alpha + \beta x$$

A simple regression equation



Quiz

Which of the following is true?

- Dependent variable is a synonym of explanatory variable
- Explanatory variable is a synonym of independent variable (Correct)
- Simple regressions sometimes use more than one variable
- None of the above

Pratice

01_Running a Regression

```
[2] import numpy as np
import pandas as pd

from scipy import stats
import statsmodels.api as sm

import matplotlib.pyplot as plt
```

```
[ ] data = pd.read_excel('D:/Python/Data_Files/Housing.xlsx')
```

```
data[['House Price', 'House Size (sq.ft.)']]
```



	House Price	House Size (sq.ft.)
0	1116000	1940
1	860000	1300
2	818400	1420
3	1000000	1680
4	640000	1270
5	1010000	1850
6	600000	1000
7	700000	1100
8	1100000	1600
9	570000	1000



data

	House Price	House Size (sq.ft.)	State	Number of Rooms	Year of Construction
0	1116000	1940	IN	8	2002
1	860000	1300	IN	5	1992
2	818400	1420	IN	6	1987
3	1000000	1680	IN	7	2000
4	640000	1270	IN	5	1995
5	1010000	1850	IN	7	1998
6	600000	1000	IN	4	2015
7	700000	1100	LA	4	2014
8	1100000	1600	LA	7	2017
9	570000	1000	NY	5	1997

Univariate Regression

```
[ ] X = data['House Size (sq.ft.)']  
    Y = data['House Price']
```

```
[ ] X
```

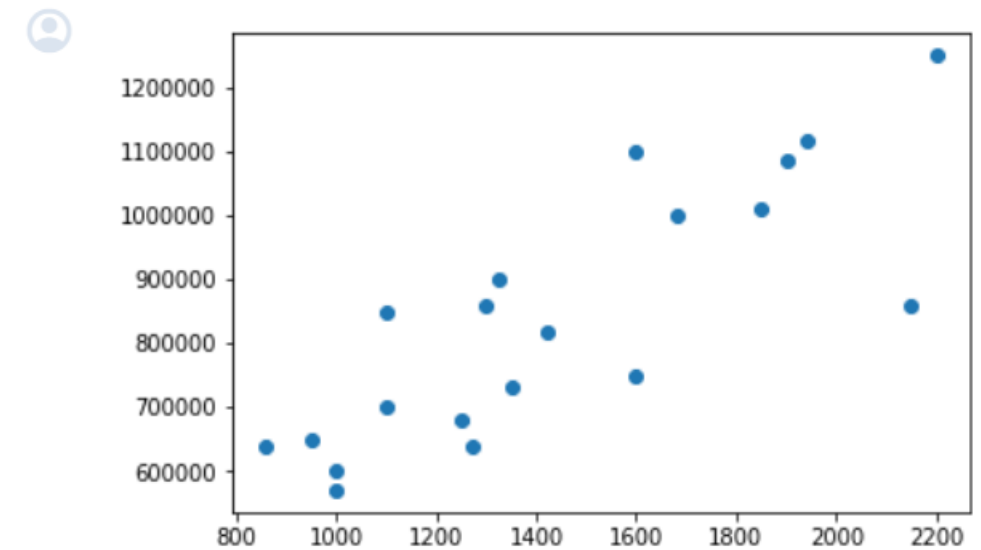
0	1940
1	1300
2	1420
3	1680
4	1270
5	1850
6	1000
7	1100
8	1600

```
[ ] Y
```

0	1116000
1	860000
2	818400
3	1000000
4	640000
5	1010000
6	600000
7	700000
8	1100000
9	570000
10	860000
11	1085000
12	1250000
13	850000
14	640000
15	900000
16	730000
17	750000
18	650000
19	680000

Name: House Price, dtype: int64

```
[ ] plt.scatter(X,Y)  
    plt.show()
```



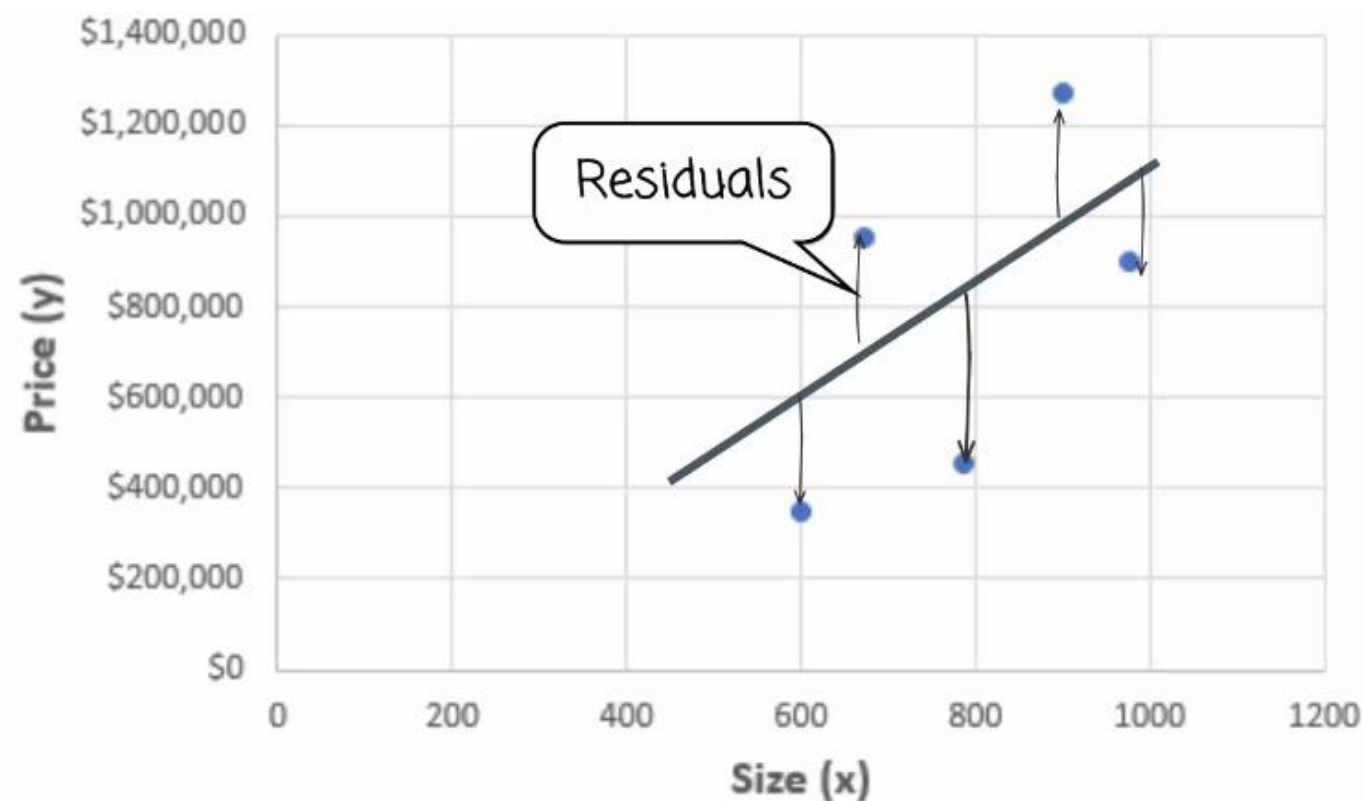
Which regressions are
reliable?

Simple Regression Coefficients

- Simple Regression Model

$$Y = \alpha + \beta x + \text{error}$$

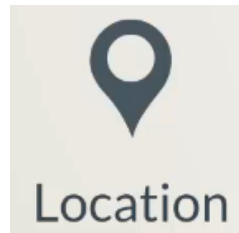
where error: residual



- The best fitting line minimizes the sum of the squared residuals.
- The coefficients found using OLS estimates (Ordinary Least Square Estimates): Alpha, Beta, Residuals

Good vs. Bad Regressions

- More than one variable determines house prices...



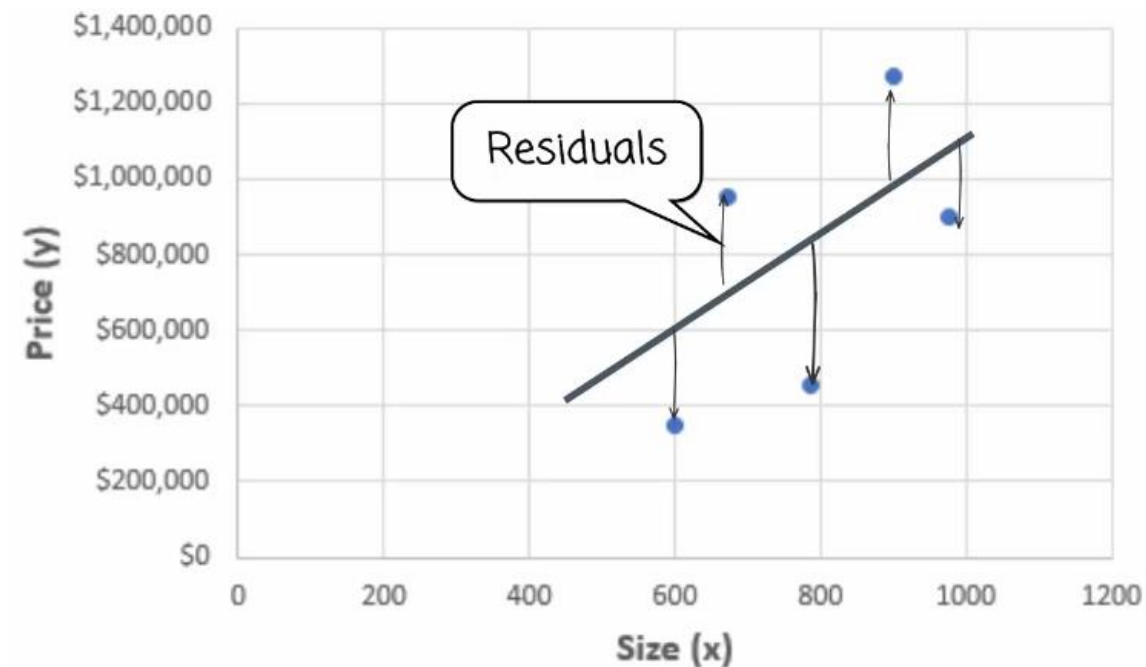
- A simple regression will omit some important factors, which will result in an estimation error.

It is useful, but it isn't perfect!

- Are all regressions created equal?
 - Certain variables are better at predicting other variables
 - Example: house size is one of the better indicators of house prices.
 - Some regression have higher explanatory power than others.
- How to evaluate Good vs. Bad Regressions?
 - Using the R square

R square

- How can we measure data dispersion and variability?



$$s^2 = \frac{\sum (x - \bar{x})^2}{n - 1}$$

$$TSS = \sum (x - \bar{x})^2$$

$$R^2 = 1 - \frac{SSR}{TSS}$$

Variance: use to measure the variability of data.

Total Sum Squares (TSS): provide a sense of the variability of data.

R Square: varies between 0% - 100%

- The higher it is, the more predictive the model has

Quiz

Which of the following values represents the slope of the regression line?

- The standard error
- Alpha
- Beta (Correct)
- R Squared

Pratice

02_Computing Alpha, Beta, and R Squared

```
import numpy as np
import pandas as pd
from scipy import stats
import statsmodels.api as sm
import matplotlib.pyplot as plt
```

```
[ ] data = pd.read_excel('D:/Python/Data_Files/Housing.xlsx')
```

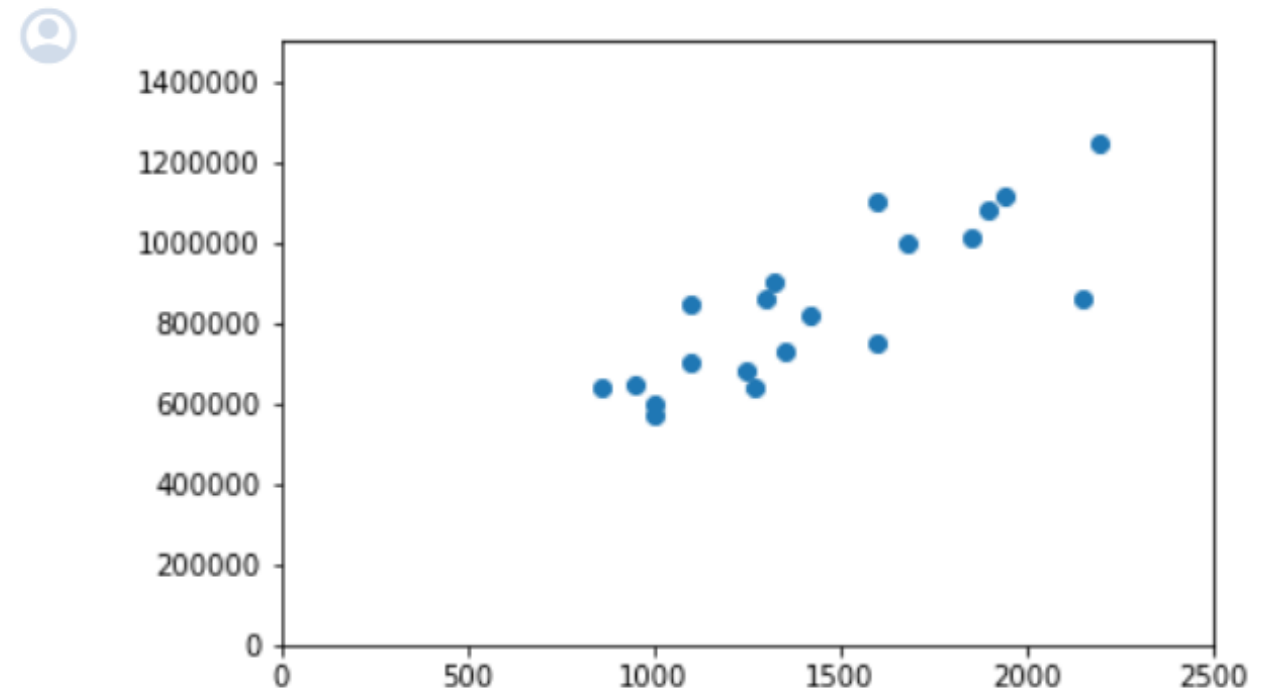
```
[ ] data
```

	House Price	House Size (sq.ft.)	State	Number of Rooms	Year of Construction
0	1116000	1940	IN	8	2002
1	860000	1300	IN	5	1992
2	818400	1420	IN	6	1987
3	1000000	1680	IN	7	2000
4	640000	1270	IN	5	1995

✓ Univariate Regression

```
▶ X = data['House Size (sq.ft.)']  
Y = data['House Price']
```

```
▶ plt.scatter(X,Y)  
plt.axis([0, 2500, 0, 1500000])  
plt.show()
```



```
[ ] X1 = sm.add_constant(X)

reg = sm.OLS(Y, X1).fit()
```

 reg.summary()



OLS Regression Results

Dep. Variable:	House Price	R-squared:	0.678
Model:	OLS	Adj. R-squared:	0.660
Method:	Least Squares	F-statistic:	37.95
Date:	Sat, 25 Mar 2017	Prob (F-statistic):	8.13e-06
Time:	13:19:31	Log-Likelihood:	-260.43
No. Observations:	20	AIC:	524.9
Df Residuals:	18	BIC:	526.8
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[95.0% Conf. Int.]
const	2.608e+05	9.76e+04	2.673	0.016	5.58e+04 4.66e+05
House Size (sq.ft.)	401.9163	65.243	6.160	0.000	264.846 538.987

Omnibus:	1.238	Durbin-Watson:	1.810
Prob(Omnibus):	0.538	Jarque-Bera (JB):	0.715
Skew:	-0.459	Prob(JB):	0.699
Kurtosis:	2.884	Cond. No.	5.66e+03

Expected value of Y:

 `260800 + 402 * 1000`

 662800

[] p_value

8.1296423772313077e-06

[] std_err

65.242995106364916

✓ Alpha, Beta, R^2:

 `slope, intercept, r_value, p_value, std_err = stats.linregress(X,Y)`

[] slope

401.91628631922595

[] intercept

260806.23605609639

[] r_value

0.82357755346969241

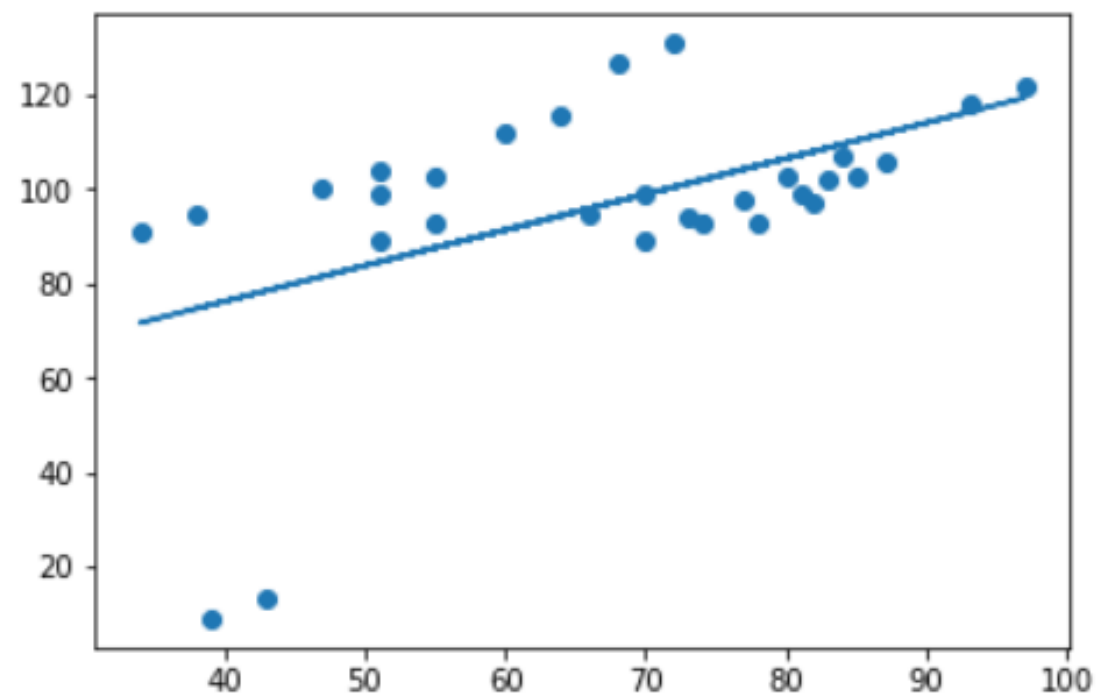
[] r_value ** 2

0.67827998657912403


```
[ ] def fitline(b):  
    return intercept + slope * b
```

```
[ ] line = fitline(X)
```

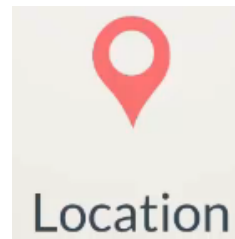
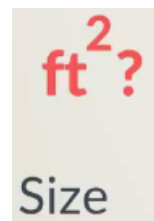
```
[ ] plt.scatter(X,Y)  
plt.plot(X,line)  
plt.show()
```



Multivariate Regression Analysis

Multivariate Regression

- Considering more explanatory variables



- These factors play a role and affect pricing?
 - There fore, a regression considering multiple variables should provide a prediction that is closer true results
 - However, we need the right variables
- Simple vs. Multivariate Regression

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_i X_i + \varepsilon_i$$

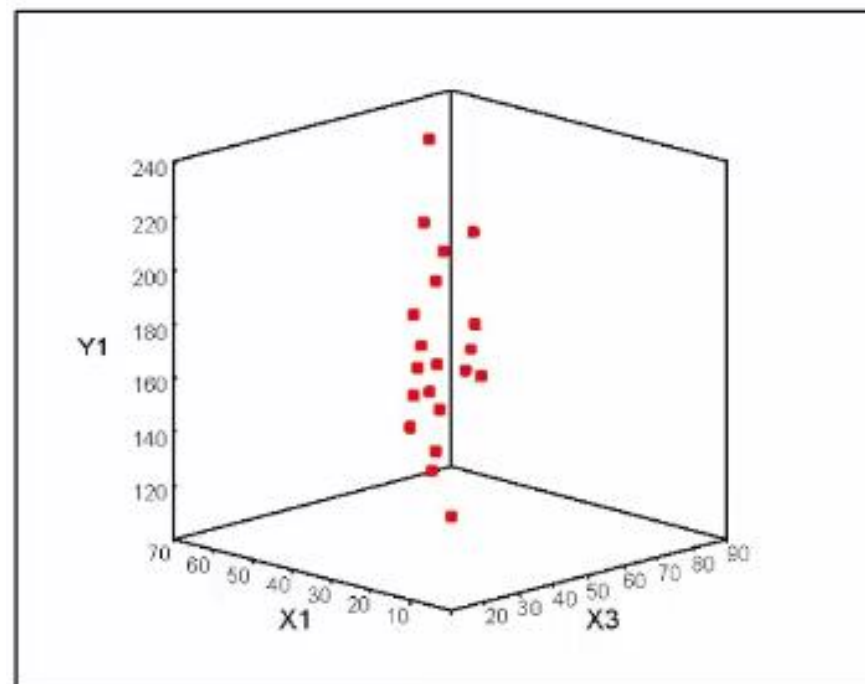
A single beta coefficient and explanatory variable

vs.

Multiple beta coefficients and explanatory variables

Multivariate Regression

- Calculating a best fitting line across multiple dimensions
- Calculation softwares facilitate regression analysis even when we use multiple variables



- **Beta** coefficients estimated in a multivariate regression can be interpreted as the marginal impact of the explanatory variable

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_i X_i + \varepsilon_i$$

$$Y_i = \beta_0 + 0.8_1 GDP \% + 1.2_2 Ind \% + \varepsilon_i$$

Multivariate Regression

R^2 How are we able to tell if an explanatory variable improves a model's predictive power?

- Helps us determine how powerful is the regression
- Between $[0, 1]$
- The higher the number of variables, the higher the R^2

Alternatively, we can compare the p-values of beta coefficients (the probability that the beta coefficients should have been different)

Low p-value: there is low chance that the beta coefficients are different than estimated

A p-value that is lower than 5% allows us to state:

We can be 95% confident that the beta coefficient we've estimated is different than 0

Quiz

How can we describe a multivariate regression in brief?

- A single beta coefficient and multiple variables without a constant
- A single beta coefficient and explanatory variable with a constant
- Multiple beta coefficients and explanatory variables with a constant (Correct)
- Consistently Higher R squared

Pratice

03_Running a multivariate regression

```

import numpy as np
import pandas as pd
from scipy import stats
import statsmodels.api as sm
import matplotlib.pyplot as plt

```

```
[ ] data = pd.read_excel('D:/Python/Data_Files/Housing.xlsx')
```

 data



	House Price	House Size (sq.ft.)	State	Number of Rooms	Year of Construction
0	1116000	1940	IN	8	2002
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5	1010000	1850	IN	7	1998
6	600000	1000	IN	4	2015
7	700000	1100	LA	4	2014
8	1100000	1600	LA	7	2017

✓ Multivariate Regression:

Independent Variables: *"House Size (sq.ft.)", "Number of Rooms", "Year of Construction"*

```
[ ] X = data[['House Size (sq.ft.)', 'Number of Rooms', 'Year of Construction']]  
    Y = data['House Price']
```

```
[ ] X1 = sm.add_constant(X)  
    reg = sm.OLS(Y, X1).fit()  
  
    reg.summary()
```

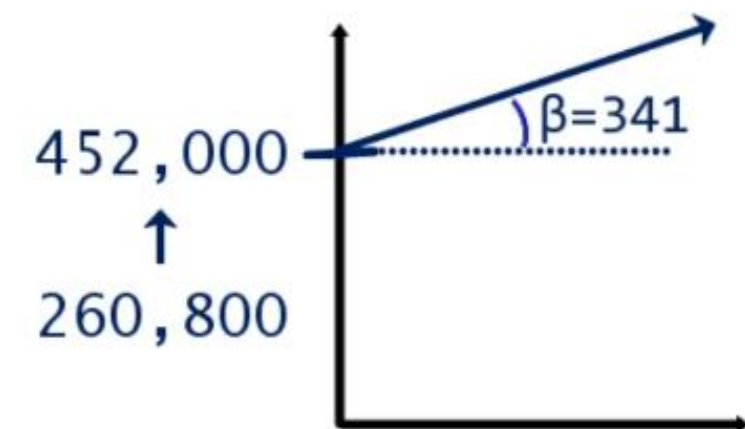
Independent Variables: *"House Size (sq.ft.)", "Number of Rooms"*

```
[ ] X = data[['House Size (sq.ft.)', 'Number of Rooms']]  
    Y = data['House Price']
```

```
[ ] X1 = sm.add_constant(X)  
    reg = sm.OLS(Y, X1).fit()  
  
    reg.summary()
```


OLS Regression Results

Dep. Variable:	House Price	R-squared:	0.736
Model:	OLS	Adj. R-squared:	0.687
Method:	Least Squares	F-statistic:	14.90
Date:	Mon, 27 Mar 2017	Prob (F-statistic):	6.82e-05
Time:	07:44:46	Log-Likelihood:	-258.43
No. Observations:	20	AIC:	524.9
Df Residuals:	16	BIC:	528.9
Df Model:	3		
Covariance Type:	nonrobust		



	coef	std err	t	P> t	[95.0% Conf. Int.]
const	-9.452e+06	5.4e+06	-1.752	0.099	-2.09e+07 1.99e+06
House Size (sq.ft.)	341.8271	179.666	1.903	0.075	-39.049 722.703
Number of Rooms	1.16e+04	3.74e+04	0.310	0.760	-6.77e+04 9.08e+04
Year of Construction	4863.5761	2697.969	1.803	0.090	-855.862 1.06e+04

>5% => not statistically significant

Independent Variables: *"Number of Rooms", "Year of Construction"*

```
[ ] X = data[['Number of Rooms', 'Year of Construction']]  
    Y = data['House Price']
```

```
[ ] X1 = sm.add_constant(X)  
    reg = sm.OLS(Y, X1).fit()  
  
    reg.summary()
```

**THANKS FOR YOUR
ATTENTION!**